STUDYHUB
CLASS 12 — MP BOARD
PHYSICS — SET 2
ORIGINAL SAMPLE QUESTION PAPER

Time: 3 Hours	Maximum Marks: 70
Medium: English	Set: 2

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.

2. Internal options are given in each question from Question No. 6 to 20.

3. Questions 1, 2 and 4 carry 6 marks each. Questions 3 and 5 carry 5 marks each. Each sub-question carries 1 mark.

4. Questions 6 to 12 carry 2 marks each. Answer within approximately 30 words.

5. Questions 13 to 16 carry 3 marks each. Answer within approximately 75 words.

6. Questions 17 to 20 carry 4 marks each. Answer within approximately 150 words.

7. Draw neat and labelled diagrams wherever necessary.

SECTION – A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct option. (1×6=6)

(i) The electric potential due to a point charge varies with distance r as:
A. r
B. 1/r
C. r^2
D. $1/r^2$

(ii) The energy stored in a capacitor is proportional to:
A. CV
B. CV^2
C. C/V
D. V/C

(iii) The SI unit of resistivity is:
A. Ω m
B. Ω/m
C. Ω m^2
D. Ω^2m

(iv) A charged particle entering a magnetic field perpendicular to it moves in:
A. A straight line
B. A circular path
C. A parabolic path
D. An elliptical path

(v) The RMS value of an alternating current having peak value I_0 is:
A. I_0
B. $2I_0$
C. $I_0/\sqrt{2}$
D. $\sqrt{2}\,I_0$

(vi) The work function of a metal is usually expressed in:
A. Ampere
B. Electron volt
C. Tesla
D. Coulomb

Q.2 Fill in the blanks. (1×6=6)

(i) The capacitance of an isolated spherical conductor is proportional to its ______.

(ii) The flow of electric charge per unit time is called ______.

(iii) The SI unit of magnetic flux is ______.

(iv) The frequency of AC supplied in India is ______ Hz.

(v) The minimum frequency required for photoelectric emission is called ______ frequency.

(vi) The nucleus of an atom contains protons and ______.

Q.3 Write True or False. (1×5=5)

(i) Electric potential is a scalar quantity.

(ii) A capacitor blocks steady direct current after it is fully charged.

(iii) The magnetic field does work on a moving charged particle.

(iv) A step-up transformer increases voltage and decreases current.

(v) Nuclear forces are approximately charge independent.

Q.4 Match Column A with Column B. (1×6=6)

Column A	Column B
(i) Electric current	(a) Tesla
(ii) Resistivity	(b) Ampere
(iii) Magnetic field	(c) Ohm metre
(iv) Magnetic flux	(d) Weber
(v) Frequency	(e) Hertz
(vi) Lens power	(f) Dioptre

Q.5 Answer in one word/one sentence. (1×5=5)

(i) What is the SI unit of electric potential?

(ii) What is the device used to measure electric current called?

(iii) Name the instrument used to detect very small electric currents.

(iv) Which phenomenon proves the wave nature of light?

(v) What is the charge of an electron?

SECTION – B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)
What is electric potential energy? Write the expression for the potential energy of two point charges.

OR
What is an electric field line? Write any one property of electric field lines.

Q.7 (2 Marks)
Two resistors of 4 Ω and 8 Ω are connected in series. Find their equivalent resistance.

OR
What is the temperature coefficient of resistance? Write its SI unit.

Q.8 (2 Marks)
State Ampere's circuital law.

OR
What is cyclotron? Write the principle on which it works.

Q.9 (2 Marks)
Define mutual inductance.

OR
What is the impedance of an AC circuit? Write its SI unit.

Q.10 (2 Marks)
What is meant by coherent sources?

OR
Write any two uses of optical fibres.

Q.11 (2 Marks)
What is de Broglie wavelength? Write its mathematical expression.

OR
What is stopping potential in the photoelectric effect?

Q.12 (2 Marks)
Define mass defect. How is it related to nuclear binding energy?

OR
What is half-life of a radioactive substance?

SECTION – C : SHORT ANSWER QUESTIONS

Q.13 (3 Marks)
Using Gauss's law, obtain the expression for the electric field due to an infinitely long uniformly charged straight wire.

OR
A charge of 4×10^{-6} C is placed in an electric field of 2×10^{4} N/C. Calculate the force acting on the charge.

Q.14 (3 Marks)
Explain the principle and working of a potentiometer. Mention one use of it.

OR
A wire of resistance 12 Ω carries a current of 2 A. Calculate the potential difference across the wire and electrical power consumed.

Q.15 (3 Marks)
Explain the principle and working of a charged particle moving in a uniform magnetic field. What determines the radius of its circular path?

OR
Explain the construction and working of a cyclotron with a neat labelled diagram.

Q.16 (3 Marks)
Derive the expression for the magnifying power of a simple microscope when the final image is formed at the least distance of distinct vision.

OR
Explain the phenomenon of polarization of light and state what it proves about the nature of light.

SECTION – D : LONG ANSWER QUESTIONS

Q.17 (4 Marks)
Derive an expression for the equivalent capacitance of capacitors connected in series.

OR
A 2 μF capacitor is charged to 100 V. Calculate: (i) Charge stored (ii) Energy stored in the capacitor.

Q.18 (4 Marks)
Explain electromagnetic induction and derive Faraday's law of electromagnetic induction.

OR
Explain the principle, construction and working of an AC transformer. Mention two sources of energy loss in a transformer.

Q.19 (4 Marks)
Derive the expression for the fringe width in Young's double-slit experiment.

OR
An object is placed 30 cm in front of a convex lens of focal length 20 cm. Using the lens formula, find the position and nature of the image.

Q.20 (4 Marks)
Explain Bohr's model of the hydrogen atom and obtain the expression for the radius of the nth orbit.

OR
Explain the principle of a semiconductor diode as a rectifier. Describe the working of a half-wave rectifier with a circuit diagram.

MARKS VERIFICATION

Questions	Marks
Q.1	6
Q.2	6
Q.3	5
Q.4	6
Q.5	5
Q.6–Q.12	14
Q.13–Q.16	12
Q.17–Q.20	16
TOTAL	70

6 + 6 + 5 + 6 + 5 + 14 + 12 + 16 = 70 Marks ✓

SHORT ANSWER KEY

Q.1: B, B, A, B, C, B. Q.2: radius; electric current; weber; 50; threshold; neutrons. Q.3: True, True, False, True, True. Q.4: (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d), (v)-(e), (vi)-(f). Q.5: Volt; ammeter; galvanometer; interference; -1.6×10^{-19} C.

Q.7: 12 Ω. Q.13: 0.08 N. Q.17: Q = 200 μC; U = 0.01 J. Q.19: image distance = 60 cm; real and inverted. For Q.6–Q.20, award marks for correct formulae, reasoning, derivation, calculation and diagrams as applicable. Full solutions are not included.